

HCopilot

Emergency Department Management System

User Manual

A practical guide for Emergency Department staff and administrators using HCopilot to manage beds, patients, staff scheduling, and department statistics.

*Université de Haute-Alsace | HCopilot v1.0
Manual generated July 2026*

Table of Contents

1. Introduction

2. Getting Started

2.1 Starting the Application

2.2 Logging In

3. Finding Your Way Around

4. Home Dashboard

5. Beds

6. Patients

7. Scheduling

8. Simulation Page

8.1 Live Clock Widget

8.2 Patient Intake Simulator

8.3 OR Scheduler (Automatic Assignment)

8.4 Staff Audit

9. Unurgent Path

10. Flow Prediction

11. Statistics

12. Settings

12.1 Beds, Doctors, Nurses & Wards

12.2 Shifts & Rotation Groups

12.3 Daily Patients & Log Patients

12.4 Datasets, Relations & Model Files

12.5 User Accounts

12.6 Reset (Danger Zone)

13. User Roles & Access Control

14. Reference: Vital Sign Ranges

15. Troubleshooting & Tips

16. Glossary

1. Introduction

HCopilot is a hospital Emergency Department (ED) management application built for the Université de Haute-Alsace. It gives ED staff a single dashboard to track beds, manage patients, schedule doctors and nurses, forecast patient flow, and monitor department performance.

With HCopilot you can:

- See which beds are free, occupied, or under repair in real time.
- Register new patients and update their clinical details as they move through the department.
- Assign patients to a bed, a doctor, and up to two nurses.
- Let the built-in scheduler automatically suggest the best bed and staff for every waiting patient.
- Simulate patient intake for training or testing purposes.
- Forecast how many patients are likely to arrive over the next 30 days.
- Review KPIs such as wait times, length of stay, triage-level breakdown, and vital sign trends.
- Manage the separate pathway for non-urgent (acuity 5) patients who do not need a bed.

■ **NOTE:** HCopilot stores all of its data in plain files on the server rather than a traditional database. This does not change how you use the application day-to-day, but it means that regular backups of the server's data folder are important.

2. Getting Started

2.1 Starting the Application

An administrator starts the HCopilot server on the hospital computer or server that will host it. Once running, the application is reached through a standard web browser — no separate installation is needed on each workstation.

Item	Detail
Application address	http://localhost:8090 (or the server's network address, e.g. http://<server-ip>:8090)
Browser requirement	Any modern web browser (Chrome, Firefox, Edge, Safari)
Login required	Yes — every user must sign in before using the dashboard

■ **NOTE:** If the page does not load, confirm with your IT administrator that the HCopilot server process is running and that your workstation can reach the server's address.

2.2 Logging In

When you open HCopilot, you will see a login screen. Enter your username and password and select **Log In**.

Username	Password	Role
admin	admin	Administrator (default account)

■ **IMPORTANT:** The default admin/admin account is created automatically the very first time the application runs. Change this password — or create named accounts for each staff member — as soon as possible. See Section 12.5 (User Accounts).

After logging in, the navigation bar only shows the pages and settings tabs your account has been granted access to (see Section 13, User Roles & Access Control). If a page you expect to see is missing, ask an administrator to review your account's permissions.

3. Finding Your Way Around

HCopilot is a single-page application: clicking a navigation item instantly swaps the visible section without reloading the browser. The main sections available (depending on your permissions) are:

Page	What it's for
Home	Dashboard with quick-access cards to jump to any section
Flow Prediction	30-day patient arrival forecast chart
Beds	Live bed grid and bed management
Patients	List of currently admitted patients; add/edit patient records
Scheduling	View and manage bed / doctor / nurse assignments
Simulation	Patient intake simulator, automatic OR scheduler, staff audit
Unurgent	Non-urgent (acuity 5) patient pathway
Statistics	Patients / Nurses / Doctors KPI dashboards
Settings	All configuration: beds, staff, shifts, groups, accounts, and more

4. Home Dashboard

The Home page is the landing screen after login. It presents quick-access cards that take you directly to the most commonly used sections (Beds, Patients, Scheduling, Simulation, Statistics). Use it as a starting point whenever you are not sure which page you need.

5. Beds

The Beds page shows a live grid of every physical bed in the department, along with an occupancy summary (total, occupied, available, under repair, and occupancy percentage).

Bed status

Status	Meaning
Available	Free and ready to receive a patient
Occupied	A patient is currently linked to this bed (calculated automatically — you never set this directly)
Under Repair	Marked out of service; takes priority over Occupied if both apply

Bed types

Type	Description
Normal	Standard ED bed
Monitor	Bed with cardiac / vital-sign monitoring equipment
ICU	Intensive care bed — required for the most critical (acuity 1–2) patients
Chariot	Temporary overflow bed, created automatically by the OR Scheduler when no ICU bed is free (see Section 8.3)

What you can do here

- Add, edit, or delete a bed (name, ward, and type).
- Toggle a bed's condition between **Available** and **Under Repair**.
- Open a bed's detail card to assign a patient to it, or to move an occupied patient to a different bed.
- Release a bed (unlink the patient without archiving their record) or fully discharge a patient from a bed (archives their stay and clears all staff links).

■ **NOTE:** Chariot beds are temporary. Once a patient leaves a chariot bed, the system automatically removes it — unless another critical patient is still waiting and there isn't enough ICU/chariot capacity to cover them, in which case it is kept until it's no longer needed.

6. Patients

The Patients page lists everyone currently admitted to the ED (active stays). From here you can:

- Add a new patient stay, entering demographics, arrival time, vital signs, pain level, triage acuity (1–5), and chief complaint.
- Edit an existing patient's details.
- Delete a patient stay (this also clears any bed/doctor/nurse links for that patient).

■ **NOTE:** The Patients page is intentionally read/edit-focused only — it does not include a "Change Bed" action. To move a patient to a different bed, use the **Beds** page (bed detail card) or **Settings** → **Daily Patients**. This avoids accidental bed moves while reviewing patient records.

A patient can only have one active stay at a time. If you try to add a patient who already has an active stay, the system will reject the duplicate entry.

7. Scheduling

The Scheduling page shows every current assignment: which bed, doctor, and (up to two) nurses are linked to each patient.

What you can do here

- View the full list of current assignments.
- Create a new assignment — link a patient to a bed and, optionally, a doctor and up to two nurses. The system checks the bed is available and the patient isn't already assigned.
- Edit an assignment — change the bed, doctor, or nurses linked to a patient.
- Remove an assignment without archiving the patient.
- Fully discharge a patient: this stamps their departure time, moves their record to the discharge archive, clears all bed/doctor/nurse links, and frees up staff capacity.

■ **NOTE:** If a patient's bed-occupation timestamp looks invalid (e.g. it predates their arrival), the system will still let you complete a discharge rather than blocking you over old data-entry mistakes.

8. Simulation Page

The Simulation page brings together three tools that share one live clock at the top of the screen: the **Patient Intake Simulator**, the **OR (automatic) Scheduler**, and the **Staff Audit**.

8.1 Live Clock Widget

A bar at the top of the page shows the current time, all shifts that are currently active (for example "Morning & Night shift" when two shifts overlap), and all rotation groups that are currently active. Dropdowns let you temporarily override the active shift or group — useful for testing how the scheduler and staff audit would behave at a different time or day, without changing the real clock.

8.2 Patient Intake Simulator

This tool is designed for training and testing. It draws a random historical patient record and lets you add it as a new active patient with one click.

- 1 Select **Draw Patient**. The system pulls a random record (excluding anyone already active) and pre-fills a new patient ID and stay ID.
- 2 Review the pre-filled clinical details in the confirmation window.
- 3 Select **Confirm** to add the patient to the active list, timestamped with the current arrival time.

8.3 OR Scheduler (Automatic Assignment)

The OR (Operations Research) Scheduler is HCopilot's core automation feature. With one click it works out the best bed, doctor, and nurses for every currently unassigned patient — those who are in the active patient list but do not yet have a bed and are not on the non-urgent path.

How to use it

- 1 Select **Run OR**. The system evaluates every unassigned patient and generates a suggestion for each.
- 2 Review each suggestion card: it shows the recommended bed, doctor, and nurse(s), plus warning flags described below.
- 3 Select **Confirm** on a suggestion to apply it — this creates the actual bed/doctor/nurse links for that patient.

How patients are prioritized — three triage lanes

A patient's ESI acuity level permanently decides which lane they are in. The lane never changes, no matter how long they wait.

Lane	Acuity	Target	Bed required
1–2 (Critical)	1, 2, or not recorded	Ward 1 (critical ward)	ICU bed (or chariot overflow)
3–4 (Standard)	3 or 4	Ward 2 / Ward 3 (balanced automatically)	Any available bed
5 (Non-urgent)	5	No bed — routed to the Unurgent path	None

■ **NOTE:** If a patient's acuity was never recorded, the scheduler treats them as acuity 1 (the most critical) so that an under-triaged patient is never accidentally deprioritized.

Within Lane 1–2, the most critical patients go first, and among equally critical patients the one who has waited longest is suggested first. Within Lane 3–4, a priority score combines acuity with waiting time so that patients who have waited a long time move up the queue; a staff member can manually raise or lower an individual patient's score before running the scheduler if there is a clinical reason to do so. Lane 5 patients are simply ordered by longest wait.

How beds are chosen

- Critical patients: the scheduler looks first for an ICU bed in Ward 1, then an ICU bed in any other ward (flagged as overflow), then any existing free chariot bed. If none is available, it flags **ICU unavailable** and asks you to either move a current ICU occupant to free up their bed, or create a new temporary chariot bed.
- Standard patients: the scheduler balances new patients between Ward 2 and Ward 3, always picking whichever currently has more free beds.

- Non-urgent patients: no bed is selected — they are routed straight to the Unurgent path.

How staff are chosen

Doctors and nurses are picked using a fairness rule: whoever currently has the fewest assigned patients is offered first; if two staff members are tied, the one who has been idle longest is preferred. Only staff on the currently active shift and rotation group, and not marked absent, are considered.

- For critical (Lane 1–2) patients, only senior doctors are offered by default; if none is on shift, the system falls back to any available doctor and flags the suggestion (**senior fallback**).
- A **Strict nurses** checkbox forces the first nurse slot to be filled by a Registered Nurse (RN) and the second by a Practical Nurse (PN); if the right role isn't available, the system falls back to any available nurse and flags it.
- Staff marked **absent** in Settings are never offered, either as a first suggestion or as a replacement.

Suggestion warning flags

Flag	What it means
Overflow	A critical patient was placed outside Ward 1 because no ICU bed was free there
Ward 1 full	Ward 1 had zero free beds of any type when this patient was processed
ICU unavailable	No ICU bed found anywhere — you must reassign an occupant or create a chariot bed
No bed available	No bed of any type could be found for this patient
Senior fallback	No senior doctor was on shift, so an intern was suggested instead
Nurse strict fallback	Strict nurse mode was on, but the required RN/PN role wasn't available
Suggest unurgent	Patient is acuity 5; no bed suggested — routed to the Unurgent path
Acuity was null	Patient had no recorded acuity — treated as acuity 1 for safety

8.4 Staff Audit

Shift changes can leave a patient linked to a doctor or nurse who has since gone off duty. The Staff Audit finds these stale assignments so they can be corrected quickly.

- 1 Select **Run Audit**. The system checks every bed-assigned patient's linked doctor and nurse(s) against the currently active shift and rotation group.
- 2 Only patients with at least one mismatched staff member are listed, with critical patients shown first and, within that, the most recently admitted first.
- 3 Each flagged patient shows which staff member is mismatched and why (wrong shift, wrong rotation group, or both), together with a ranked list of replacement candidates.
- 4 Pick a replacement from the dropdown and select **Apply swap**. The assignment updates immediately and both staff members' patient counts are adjusted.

Replacement candidates are always split into two groups, shown in this order: staff who are currently on duty (ranked by fewest patients, then longest idle) and, as a fallback, staff who are not currently on shift/group but are not marked absent. Absent staff never appear as candidates. The same **Strict nurses** option available in the OR Scheduler can also be applied here.

9. Unurgent Path

Acuity 5 (non-urgent) patients are treated without occupying a physical bed. The Unurgent page lists every patient currently on this pathway, along with their linked doctor and nurse(s).

Patients enter this pathway automatically: when the OR Scheduler suggests a non-urgent routing for an acuity-5 patient and a staff member confirms it, the patient is flagged as unurgent and linked to staff, but no bed is assigned.

Discharging an unurgent patient

- 1 Select **Discharge** for the patient.
- 2 Confirm the departure time (defaults to now; it must be after the arrival time).
- 3 The system archives the stay, removes the patient from the active list, and clears their doctor/nurse links (no bed needs to be released, since none was ever assigned).

10. Flow Prediction

This page forecasts how many patients are likely to arrive at the ED over the coming days, using a machine-learning model trained on historical arrivals and weather data.

- The forecast chart shows up to 30 days ahead.
- A historical chart shows actual daily arrival counts for the recent past for comparison.
- Summary statistics for the full historical dataset are shown alongside the charts.

■ **NOTE:** The very first time you open this page, the model and historical data are loaded from disk, which can take a moment. Subsequent visits are much faster, since the data stays cached until it changes.

The forecast is generated one day at a time: each predicted day feeds into the calculation for the next day, alongside calendar information (day of week, month) and a temperature estimate based on recent averages. This means the forecast is most reliable for the first week and gradually more approximate further out.

11. Statistics

The Statistics page is a three-tab dashboard — **Patients**, **Nurses**, and **Doctors** — built from your active and archived patient records and current staff rosters. Which tabs you can see depends on your account's permissions; if you can't see the Patients tab, the page opens on the first tab you do have access to.

Patients tab

Seven panels covering: wait-to-bed time distribution, length-of-stay distribution, triage (ESI 1–5) breakdown, hourly and daily arrival throughput, top chief complaints, vital sign aggregates, and a data-quality warning banner (shown

only when data quality drops below 80%).

■ **NOTE:** Extreme values are automatically excluded as likely data-entry errors: wait times over 24 hours and lengths of stay over 7 days are left out of the averages and charts.

Nurses tab / Doctors tab

Both tabs share the same layout:

- Headline KPIs: total headcount, active (non-absent) count, absence rate, average patient load, and unique patients covered.
- A role/type distribution chart (nurse roles: RN, PN, Bed Admission; doctor types: attending vs. intern).
- A shift distribution chart showing how many staff are on each shift.
- A patient-load table grouping staff by how many patients they currently carry (0, 1, 2–3, 4–5, or more than 5).
- A staff lookup: pick any staff member to see their profile, current patients (with acuity and chief complaint), assigned beds, and a summary of which wards they are covering.

Vital sign "normal" ranges used for flags

Vital sign	Normal range	Unit
Temperature	36.1 – 37.5	°C
Heart rate	60 – 100	bpm
Respiratory rate	12 – 20	breaths/min
Oxygen saturation	95 – 100	%
Systolic blood pressure	90 – 140	mmHg
Diastolic blood pressure	60 – 90	mmHg

12. Settings

The Settings page brings together every configuration tool in HCopilot. It is organized into sub-tabs; which tabs you see depends on your account's permissions (see Section 13).

12.1 Beds, Doctors, Nurses & Wards

- **Beds** — add, edit, or delete beds; assign each bed to a ward.
- **Doctors** — add, edit, or delete doctor/intern records; toggle a doctor's absent status without deleting their record.
- **Nurses** — same as Doctors, with an additional role field (RN, PN, or Bed Admission).
- **Wards** — add, edit, or delete wards. Deleting a ward automatically clears any beds, doctors, or nurses linked to it.

■ **IMPORTANT:** Ward 1 is the department's critical ward. The OR Scheduler always routes the most critical patients (Lane 1–2) there specifically — do not repurpose or rename Ward 1 to a non-critical role.

12.2 Shifts & Rotation Groups

Shifts define named time windows (e.g. "morning", "night") using a start and end hour in 24-hour format. The two default shifts are:

Shift	Hours
morning	07:00 – 19:00
night	19:00 – 07:00

To add a custom shift, provide a name and start/end hour. For a shift that crosses midnight (such as a night shift), set the end hour lower than the start hour — the system handles this automatically. Shift names must be unique. If you rename a shift, every doctor and nurse who referenced the old name is updated automatically, so existing assignments keep working.

Rotation groups map a set of weekdays to a set of staff. The two default groups are:

Group	Days covered
Group 1	Monday – Thursday
Group 2	Friday – Sunday

Days are numbered 0 (Monday) through 6 (Sunday). As with shifts, group names must be unique, and renaming a group automatically updates every doctor and nurse who referenced it.

■ **NOTE:** After changing a shift or group's underlying time window or days, double-check that every doctor and nurse still has the correct shift/group selected in their own record.

12.3 Daily Patients & Log Patients

- **Daily Patients** — a direct table view of every active stay, with add/edit/delete access. This is also where you can reassign a patient's bed outside of the Beds page.
- **Log Patients** — a read/edit/delete view of discharged (archived) stays, useful for correcting historical records.

12.4 Datasets, Relations & Model Files

- **Datasets** — a paginated browser for the underlying data files, including large historical datasets. An optional **Refresh** action can pull fresh data from an upstream server if one is configured; the system works fully without it.
- **Relations** — a generic viewer for the underlying links between patients, beds, doctors, nurses, and wards. Mainly useful for troubleshooting.

- **Models** — lists the trained forecasting model file(s) used by Flow Prediction, along with file size and last-updated date.
- **Features** — feature-flag style toggles for configuring optional behavior.

12.5 User Accounts

Administrators manage login accounts from Settings → Accounts.

- Each account has a role of **admin** (full access) or **user** (limited to what is explicitly granted).
- For a **user** account, choose which pages, which Settings sub-tabs, and which Statistics tabs they can see.
- Passwords are stored securely (hashed) — nobody, including administrators, can retrieve a forgotten password. Reset it instead.
- The system will not let you delete the last remaining admin account, to prevent locking everyone out.

■ **IMPORTANT:** Access restrictions only control what is shown in the browser. Anyone with direct network access to the server could still reach the underlying data through the API. Ask your IT administrator to make sure the server is only reachable from trusted hospital networks.

12.6 Reset (Danger Zone)

The Reset tab wipes selected data back to empty. Use it only for starting a fresh environment (for example, before a training session) — never on a live system with real patient data unless you are certain.

Reset option	Wipes
Patients	All active and archived patient stays, and their bed/doctor/nurse links
Beds	All beds and their ward/patient links
Doctors	All doctor records and their links
Nurses	All nurse records and their links
Wards	All wards and their bed/doctor/nurse links
Relations	All patient/ward links between beds, doctors, and nurses
Full reset	Everything listed above

■ **IMPORTANT:** Every reset option is irreversible. Always confirm with your team before using this tab, and make sure a backup of the data folder exists beforehand.

13. User Roles & Access Control

Role	Description
admin	Full access to every page, every Settings sub-tab, and every Statistics tab
user	Access limited to whatever pages, Settings sub-tabs, and Statistics tabs an administrator has explicitly granted

If a **user** account has access to Statistics but no specific Statistics tabs were chosen, all three tabs (Patients, Nurses, Doctors) are shown by default.

If a page or settings tab you need is missing from your navigation bar, contact an administrator to review your account's permissions from Settings → Accounts.

14. Reference: Vital Sign Ranges

Every vital sign entered anywhere in HCopilot (Patients page, Settings → Daily/Log Patients, Simulation) is checked against these acceptable ranges. A value outside the range will be rejected so you can correct it before saving.

Field	Minimum	Maximum	Unit
Temperature	26	46	°C
Heart rate	20	300	bpm
Respiratory rate	4	100	breaths/min
Oxygen saturation	0	100	%
Systolic BP	40	300	mmHg
Diastolic BP	20	200	mmHg
Acuity (ESI level)	1	5	—

15. Troubleshooting & Tips

Situation	What to do
Can't log in	Confirm your username/password with an administrator. Accounts are never restored automatically if the password is forgotten — it must be reset.
A page or tab is missing	Your account may not have been granted access. Ask an administrator to check Settings → Accounts.
OR Scheduler shows "ICU unavailable"	Either move a current ICU patient to a normal/monitor bed to free capacity, or confirm the suggestion with the chariot-bed option to create a temporary overflow bed.
A vital sign won't save	Check it falls inside the accepted range in Section 14 — values outside these ranges are rejected.

Situation	What to do
Staff Audit keeps flagging the same person	Their shift or rotation-group field in Settings → Doctors/Nurses may not match any currently active shift or group — review and correct it, or apply a swap instead.
A discharge is blocked	The system checks that the departure time is after the arrival time. Correct the timestamps and try again.
Flow Prediction feels slow the first time you open it	This is expected — the model and historical data load from disk on first use, then stay cached for speed.
Application won't load at all	Ask your administrator to confirm the HCopilot server process is running and reachable from your network.

16. Glossary

Term	Meaning
Acuity / ESI level	Emergency Severity Index triage level, 1 (most critical) to 5 (least urgent)
Chariot bed	A temporary overflow bed created automatically when no ICU bed is free
LOS	Length of Stay — time from bed assignment to discharge
OR Scheduler	The automatic tool that suggests bed, doctor, and nurse assignments
RN / PN	Registered Nurse / Practical Nurse
Rotation group	A named set of weekdays used to decide which staff are "on duty" that day
Shift	A named time window (e.g. morning, night) assigned to staff
Staff Audit	The tool that finds patients linked to staff who are no longer on duty
Unurgent path	The pathway for acuity-5 patients who are treated without a bed
Ward 1	The department's designated critical-care ward for Lane 1–2 patients